

Jason Yi

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EDUCATION

University of North Carolina at Chapel Hill

Bachelor of Science in Computer Science, Statistics

Chapel Hill, NC

Exp Grad: Dec 2027

- **TA:** Algorithms (F25, S25), System Fundamentals in C (F24), Data Structures and Analysis in Java (F23, S24)

EXPERIENCE

Snowflake

August 2026 - December 2026

Software Engineer Intern

Menlo Park, CA

- Optimized **Cortex Agent Memory (CAM)** search in **Go**, profiling with **pprof** against calibrated benchmarks and shipping compiled CEL filter caches and Arrow columnar layouts without regressing retrieval quality
- Benchmarked **4-bit TurboQuant** vs. **uint8 quantization** on search cost, storage, and recall; built batch Search, Get, and Delete APIs that collapsed client **RPC** fan-out for upstream Cortex Sense workloads

Google

May 2026 - August 2026

Software Engineer Intern

New York, NY

- Migrated the **Fuchsia OS Bluetooth** ATT stack to a memory-safe **async Rust** architecture, eliminating legacy **C++** callback-driven heap overhead by engineering a statically dispatched, zero-allocation database registry
- Developed the **Rust** code generation backend for the **Emboss** compiler toolchain, implementing compile-time typestate tracking to eliminate virtual dispatch and ensure zero-overhead, memory-safe data parsing

Amazon Web Services

May 2025 – August 2025

Software Development Engineer Intern

Seattle, WA

- Built an optimized test suite for **AWS Compute Optimizer** in **Kotlin** and **Python** with **AWS AppConfig**, **AWS Lambda**, and **S3** to reduce false **Sev-2** alerts by decoupling canary tests from upstream data
- Accelerated bug bash validation by **20%** by integrating a **GenAI**-driven agentic workflow to automate ingestion of schema-validated mock data into existing indexing pipelines to streamline testing and reduce operation delays

Fidelity Investments

June 2024 – August 2024

Software Engineer Intern

Durham, NC

- Developed Backend services in **GraphQL** for **Account Opening** impacting **50+ million users**
- Implemented customer info, address validation, and risk analysis services to prevent user fraud or illegal activity during account opening using **TypeScript** and **GraphQL** by matching data from multiple downstream APIs

PROJECTS

Synapse: Distributed AI Scheduler 🐧 | Go, gRPC, Protocol Buffers, Linux Kernel

- Engineered a fault-tolerant **Go** scheduler implementing **Gang Scheduling** to enforce atomic resource allocation across dynamic worker nodes, eliminating deadlocks common in **distributed multi-GPU training** workloads
- Developed a bi-directional **gRPC** control plane to orchestrate a custom **Rust** container runtime (**Carapace**), bridging high-level scheduling with low-level **Linux syscall** execution via the **Fork/Exec** model

Carapace: Linux Container Runtime 🐧 | Rust, C++, Linux Syscalls, Cgroups, Namespaces, Bash

- Architected a container runtime in **Rust** that interfaces directly with the **Linux Kernel** via **syscalls**, leveraging **Namespaces** (UTS, PID, Mount) and **chroot jails** to achieve strict process isolation
- Enforced resource quotas via **Control Groups (Cgroups v2)** to prevent process exhaustion attacks (e.g., fork bombs), and developed a **C++ FFI** interoperability layer to bridge Rust with low-level system inspection tools

AutoReturns 🐧 | Python, PyTorch, Scikit-learn, NumPy, Pandas, Seaborn, Matplotlib

- Analyzed **30M+** rows of U.S. stock return data (**CRSP**, **S&P 500**, **Russell 3000**) to document stylized facts (e.g., volatility clustering, fat tails) using **PCA**, **autocorrelation**, and **return histograms**
- Reduced dimensionality of stock return data by **90%** using kernel PCA in **scikit-learn** and autoencoders in **PyTorch**, preserving **≥85%** variance and uncovering latent relationships across asset classes

TECHNICAL SKILLS

Languages: C/C++, Go, Python, Rust, Java, Kotlin, TypeScript, JavaScript, HTML/CSS, SQL, Swift/SwiftUI

Frameworks/Libraries: PyTorch, Tensorflow, NumPy, Pandas, Matplotlib, React.js, GraphQL, Node.js

Developer Tools: AWS, Kubernetes, Git, GitHub, Linux Kernel, Vim, Jira, Jenkins, XCode